

# FINAL INTERNSHIP PROJECT REPORT

## Context Anchored Question Answering System using Prompt Engineering and Google Gemini

**Student:** Pradeep Kumar  
**Project Type:** Internship Project

### Abstract

This project develops a Flask-based Question Answering system that answers questions only from a supplied reference document. Prompt Engineering, context anchoring, keyword retrieval, and response validation are used to reduce hallucinations.

### Introduction

Large Language Models are powerful but may generate incorrect answers without context. This project demonstrates a closed-book QA system that restricts the model to the supplied document.

### Problem Statement

General-purpose AI models may hallucinate. The objective is to answer only from the uploaded document and return 'Information Not Found' when the answer is unavailable.

### Objectives

Develop a Flask web application, integrate Google Gemini, retrieve the relevant paragraph, generate context-aware responses, validate outputs, and display source citations.

### Technologies

Python, Flask, Google Gemini API, HTML, CSS, Prompt Engineering.

### System Architecture

User → Flask → Document Loader → Retriever → Prompt Builder → Gemini API → Validator → Browser.

### Module Description

**app.py:** Main controller.  
**document\_loader.py:** Loads and splits the document.  
**retriever.py:** Finds the best matching paragraph.  
**prompts.py:** Builds a context-anchored prompt.  
**gemini\_client.py:** Communicates with Gemini.  
**validator.py:** Validates AI responses.

### Workflow

1. User enters a question.
2. Document is loaded.
3. Relevant paragraph is retrieved.
4. Prompt is generated.
5. Gemini produces an answer.
6. Validator checks the response.

7. Answer and source are displayed.

Features

Closed-book QA, context anchoring, prompt engineering, keyword retrieval, response validation, source citation, modular architecture, responsive web UI.

Advantages

Reduces hallucinations, simple architecture, modular implementation, easy maintenance, user-friendly interface.

Limitations

Supports text documents only, keyword-based retrieval, depends on Gemini API availability and quota.

Future Scope

PDF support, semantic search, embeddings, vector databases (FAISS/ChromaDB), multi-document support, chat memory, deployment.

Testing

Representative test cases include office timings, leave policy, remote work, dress code, and invalid questions returning 'Information Not Found'.

Conclusion

The project demonstrates how prompt engineering and context anchoring can improve the reliability of AI-assisted question answering while maintaining a clean and modular architecture.

References

Python Documentation, Flask Documentation, Google Gemini Documentation.

Technology Summary

| Component  | Purpose                      |
|------------|------------------------------|
| Flask      | Web framework                |
| Gemini API | Answer generation            |
| Retriever  | Relevant paragraph selection |
| Validator  | Response validation          |
| HTML/CSS   | User interface               |